

Execution-order comparison

Pipeline: Normal LLM | 2 runs | 2 identical question texts | Reference: R1

Reference selected automatically: its actual execution sequence matches the selected questions in questionbank.xlsx sheet/row order.

Run	Folder / model
R1	exp1_llm_only_2026-10-09_19-37-53 llama3
R2	exp1_llm_only_2026-10-09_19-42-08 llama3

Main observations

These are observed differences, not a causal estimate of the effect of shuffling. Output length, background activity and measurement variation may contribute.

Recorded model names match; this does not establish identical model versions or generation settings.

Generation settings, token counts and RAG index identity are not consistently recorded by these reports; equality of those conditions cannot be verified.

Lowest observed emissions: R2, 1.1159 g CO₂e. Spread between highest and lowest: 0.12103 g (10.846% relative to the lowest).

R2 versus R1: identical execution sequence. 2 valid successful question pairs; 2 lower, 0 higher, 0 unchanged emissions.

Largest absolute change: Q1, -0.066311 g; execution position 1 to 1. Net change over these matched successful pairs: -0.12103 g.

R2: energy_wh change summed over 2 matched successful pairs: -0.16964.

R2: latency_s change summed over 2 matched successful pairs: -16.934.

Recorded metadata differences: Ordering, Seed, Runtime (s). Unknown or unrecorded values do not establish equivalence.

Historical zero defaults cannot be distinguished from measured zeros. Missing measurements are excluded and partial sums labelled. Percentage changes with a zero reference are N/A.

Recorded shuffling schemes

These rules and seeds come from saved ordering metadata, with CSV metadata used when available. Missing rules are not inferred from the observed sequence. A shuffled scheme can sometimes produce the same sequence.

Run	Recorded scheme	Seed
R1 (reference)	Keep original order	Not applicable
R2	Shuffle question types; keep questions within each question type in original order	182766017

Execution sequence comparison

Positions 1-2. Reference: R1. Read down a column to follow that run. Read across a row to compare what ran at the same position.

Up arrow / blue: Earlier by N. Down arrow / amber: Later by N. Equals / grey: Same position. N counts positions relative to the reference. Colours show movement only.

Position	R1 - Reference Keep original order Seed: Not applicable	R2 Shuffle question types; keep questions within each question type in original order Seed: 182766017
1	= Q1 Reference	= Q1 Same position
2	= Q2 Reference	= Q2 Same position

Run cost and coverage

Observed sums include failed requests. Coverage is valid readings / all questions. Partial sums cannot be treated as complete batch costs.

Run	Emissions (g)	Energy (Wh)	Request time (s)	Failures / unknown
R1	1.2369 [2/2]	1.7337 [2/2]	208.42 [2/2]	0
R2	1.1159 [2/2]	1.5641 [2/2]	191.49 [2/2]	0

Batch costs: R1-R2

Emissions (g CO2e)

Observed sums across all requests; partial coverage is labelled.

R1		1.237
	Valid 2/2	
R2		1.116
	Valid 2/2	

Energy (Wh)

Observed sums across all requests; partial coverage is labelled.

R1		1.734
	Valid 2/2	
R2		1.564
	Valid 2/2	

Metadata differences

Different recorded values are marked *. Not recorded means unavailable, not equal.

Setting	R1	R2
Folder *	exp1_llm_only_2026-10-09_19-37-53	exp1_llm_only_2026-10-09_19-42-08
Model(s)	llama3	llama3
Pipeline	Normal LLM	Normal LLM
Ordering *	Keep original order	Shuffle question types; keep questions within each question type in original order
Seed *	Not recorded	182766017
Questions	2	2
Runtime (s) *	208.71036314964294	191.53168296813965
Completion evidence	Summary present	Summary present
codecarbon_version	"3.3.0"	"3.3.0"
os	"Windows-11-10.0.26200-SP0"	"Windows-11-10.0.26200-SP0"
effective_scope	"machine"	"machine"
measure_power_secs	1	1
pue	1.0	1.0
country_iso_code	"IND"	"IND"
config_profile	"inference"	"inference"
configured_settings	{"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}	{"allow_multiple_runs": true, "country_iso_code": "IND", "log_level": "error", "measure_power_secs": 1, "pue": 1.0, "save_to_api": false, "tracking_mode": "machine"}
hardware	{"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_count": 0, "gpu_model": "", "ram_total_size": 15.69796371459961}	{"cpu_model": "12th Gen Intel(R) Core(TM) i5-1235U", "gpu_count": 0, "gpu_model": "", "ram_total_size": 15.69796371459961}
backends	{"cpu": "unknown / not exposed by this version", "gpu": "unknown / not exposed by this version", "ram": "unknown / not exposed by this version"}	{"cpu": "unknown / not exposed by this version", "gpu": "unknown / not exposed by this version", "ram": "unknown / not exposed by this version"}
hardware_evidence	[{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]	[{"class": "RAM", "mode": "unknown"}, {"class": "CPU", "mode": "windows_emi"}]

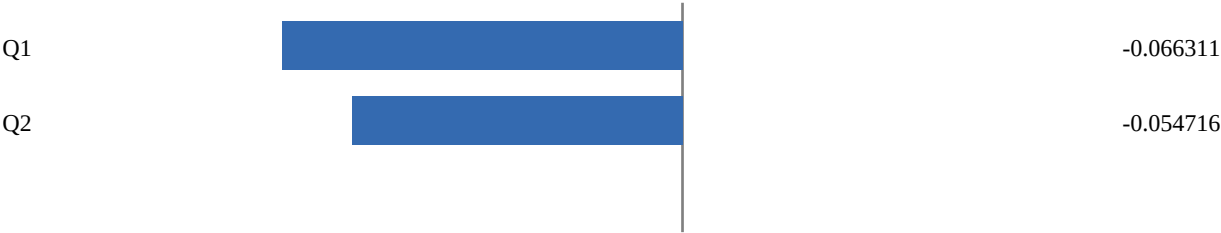
Question emissions change (%)

Relative to R1. Blue = lower, red = higher; grey = missing, failed, or zero reference. Q labels map to exact question text in the comparison table. Color saturates at +/-100%; cell numbers retain the actual change.

	R1	R2
Q1	0	-11.961
Q2	0	-8.0173

Largest changes: R2 versus R1

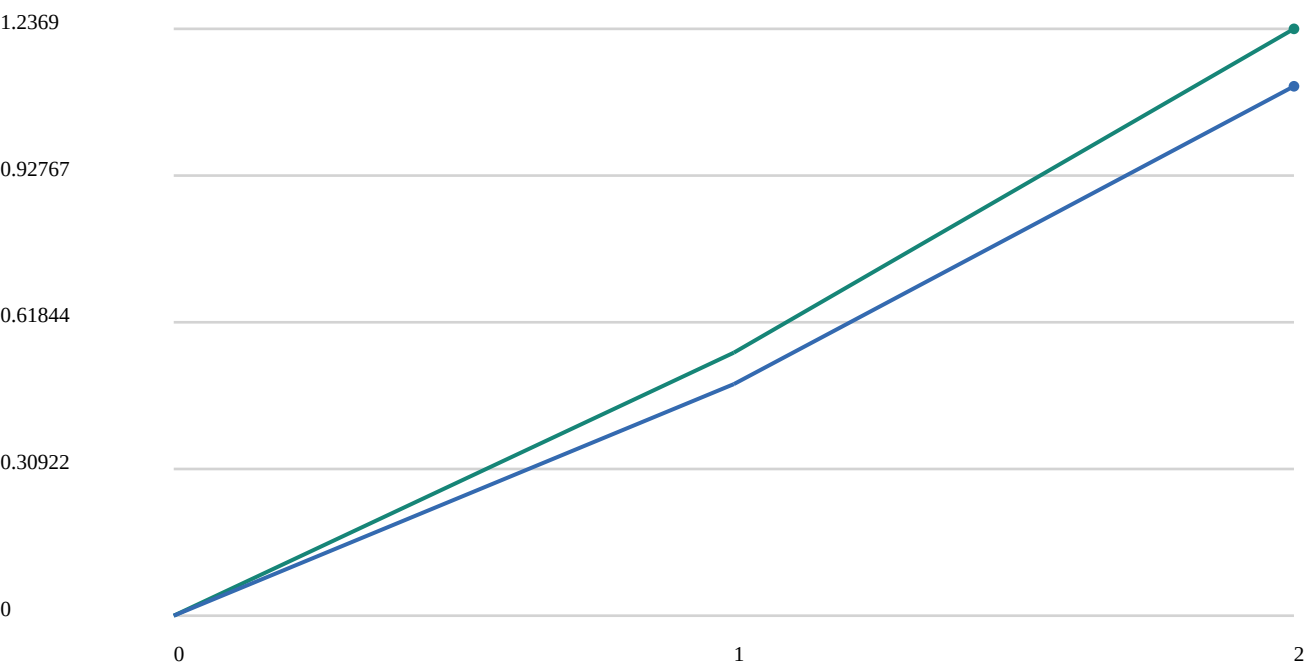
Absolute emissions changes for up to 12 matched successful questions. Left = decrease; right = increase. Labels show g CO2e.



Cumulative emissions through execution

X: number of questions executed. Y: cumulative g CO2e. Question identities at a given position can differ. Curves stop at the first missing measurement rather than implying zero cost. Failed-request measurements are included.

R1: 2/2 questions accumulated
R2: 2/2 questions accumulated



Aligned question comparison

Q identifiers follow the reference run. Text matching is exact. Mean emissions are calculated per question across all selected runs, using recorded values only; missing readings are excluded and the contributor count is shown. Recorded costs for failed attempts are included. Percentage changes are unavailable for a zero reference.

Q1: Repeated trials of the same operational task can cause agent pass rates to drop — propose a consistency-improvement approach appropriate for a local, resource-constrained LLM agent operating without external APIs.

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	1	ok	0.55441	0.52125 (2/2)	0.77709	92.024	0	0
R2	1	ok	0.48809	0.52125 (2/2)	0.68414	82.275	-0.066311	-11.961

Q2: In a customer support setting, how should a business weigh the tradeoff between utilizing a heavy multi-agent LLM workflow that yields higher resolution accuracy, versus the reduced operational compute load of a smaller, less accurate standalone model?

Run	Position	Status	g CO2e	Mean g (n/runs)	Wh	Seconds	Delta g	Change %
R1	2	ok	0.68248	0.65512 (2/2)	0.95661	116.4	0	0
R2	2	ok	0.62777	0.65512 (2/2)	0.87991	109.22	-0.054716	-8.0173

Question label key

Q labels follow the execution order in R1. The same exact question text has the same label in every run and throughout this PDF.

Label	Exact question text
Q1	Repeated trials of the same operational task can cause agent pass rates to drop — propose a consistency-improvement approach appropriate for a local, resource-constrained LLM agent operating without external APIs.
Q2	In a customer support setting, how should a business weigh the tradeoff between utilizing a heavy multi-agent LLM workflow that yields higher resolution accuracy, versus the reduced operational compute load of a smaller, less accurate standalone model?